The Exact Sidon Constant of Six Consecutive Frequencies

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Abstract

We determine the Sidon constant of $\{0,1,2,3,4,5\}$ and classify all extremizing complex polynomials. The sharp constant is a specified root of a quartic equation; complex coefficients give no advantage over real ones. The proof combines an exact algebraic extremizer, a locally sharp Toeplitz majorant with a degenerate Schur complement, and a finite certified covering of the remaining phase space. All finite computations are verified by an accompanying integer-arithmetic certificate.

1 Introduction

For a finite subset $E \subset \mathbb{Z}$, its Fourier-analytic Sidon constant is the least C such that

$$\sum_{k\in E} |a_k| \le C \sup_{|z|=1} \left| \sum_{k\in E} a_k z^k \right| \qquad (a_k \in \mathbb{C}). \tag{1}$$

We consider the initial segments $E = \{0,\dots,n\}$, writing S_n for this constant. Even for such small spectra, identifying the exact constants requires more than Parseval's inequality. Indeed, Parseval gives $S_n \le \sqrt{n+1}$, while sampling at a suitable rotation of the nth roots of unity improves this to $S_n \le \sqrt{n}$ for $n \ge 1$. The latter bound, historically associated with Newman and Shapiro, is attained at $n = 1, 2, 4$; see the historical discussion in [6, 5]. For large n, the theory of unimodular and ultraflat polynomials explains the asymptotic sharpness of the square-root scale [4, 2]. These asymptotic constructions, however, do not determine a prescribed finite-degree optimum.

The degree-three complex problem was recently resolved by Yu, who established $S_3 = 5/3$ with a formalized proof [6]; Neuwirth's discussion [5] contains earlier structural perspectives and examples. The next odd degree is particularly natural because an optimal degree-five polynomial is subject to a six-dimensional coefficient phase problem with genuine complex degrees of freedom. We settle this problem and determine all cases of equality.

Let w be the unique real root in $(1/8, 13/100)$ of

$$f(w) = 16w^4 + 10w^3 - 8w + 1 = 0. \tag{2}$$

Define the six real numbers $p_0,\dots,p_5$ by

$$p_k = \frac{A_k + B_k w + C_k w^2 + D_k w^3}{25875}, \tag{3}$$

where

k	A_k	B_k	C_k	D_k
0	3112	4752	9708	-9248
1	9870	-6480	-28920	-12480
2	13460	-4590	-34860	-31840
3	7780	11880	24270	-23120
4	-15705	-2430	32280	64320
5	7358	-3132	-2478	12368

Set $P_*(z) = \sum_{k=0}^{5} p_k z^k$ and

$$\Lambda = \frac{3819 + 324w - 4304w^2 - 8576w^3}{1725}. \tag{4}$$

Theorem 1.1 (Sharp constant and all equality cases)**.** *For every polynomial $P(z) = \sum_{k=0}^{5} a_k z^k$ with complex coefficients,*

$$\sum_{k=0}^{5} |a_k| \le \Lambda \max_{|z|=1} |P(z)|, \tag{5}$$

where

$$\Lambda = 2.186529771892569\ldots$$

is the unique root in $(2.18, 2.19)$ *of*

$$675X^4 - 1026X^3 - 936X^2 - 414X + 677 = 0. \tag{6}$$

Equality for a nonzero polynomial holds if and only if, after normalizing its supremum norm to one, it has one of the forms

$$\gamma P_*(\omega z), \qquad \gamma z^5 P_*(\omega/z), \qquad |\gamma| = |\omega| = 1. \tag{7}$$

Consequently $S_5 = S_5^{\mathbb{R}} = \Lambda$, where $S_5^{\mathbb{R}}$ denotes the analogous constant restricted to real coefficients.

The proof is computer-assisted in a precise, limited sense. All analytic reductions, the local matrix construction, and the implication from finite certificates to the universal inequality are proved here. A companion program `Sidon5_Exact_Closure_Certificate_Revised.py` verifies explicitly specified identities in the quartic field $\mathbb{Q}(w)$ and a complete finite set of positive-definiteness inequalities using outward integer intervals. The embedded finite data are part of the proof, not samples of an optimization computation. The certificate was executed successfully with Python 3's standard library and needs no network access or numerical solver.

The organizing idea. Writing the coefficient ℓ^1 norm as the maximum of a family of linear functionals transforms the question into an optimization over phases. Positive Toeplitz matrices then provide dual upper bounds for the norm of each functional. A sharp dual matrix at a known extremizer is necessarily singular; ordinary continuity cannot preserve a bound with the exact constant. We therefore correct the Toeplitz matrix through cubic order, resolve the singular directions by a Schur complement, and obtain a genuinely sharp bound in a fixed neighborhood. Away from these neighborhoods, strict nonsharp bounds suffice. Crucially, an inequality that exploits alignment of the coefficient phases excludes whole boxes from containing a global maximizer, without trying to bound every functional inside each box sharply. This division between local sharpness and global exclusion is the mechanism behind the proof.

The philosophy of locally strong certificates versus global approximation has parallels in the sum-of-squares hierarchy for trigonometric optimization [1]; matrix-order norm descriptions appear in related form in [3]. Neither of these papers is used as an external theorem below: the required Toeplitz and Schur-complement arguments are proved explicitly.

2 An exact algebraic extremizer

The polynomial (2) has opposite signs at $1/8$ and $13/100$. Its derivative $64w^3 + 30w^2 - 8$ is negative throughout this interval, hence w is uniquely defined. The companion certificate uses rational bisection to enclose w and checks that $p_0, p_1, p_2, p_3, p_5 > 0$ and $p_4 < 0$. Thus with

$$s = (1, 1, 1, 1, -1, 1)^T, \tag{8}$$

we have $s^T p = \sum |p_k| = \Lambda$. The program also verifies $P_*(1) = 1$ and the quartic identity (6), by reduction modulo $f(w)$ in $\mathbb{Q}[w]$; uniqueness of the root in $(2.18, 2.19)$ follows because the polynomial in (6) is strictly increasing there and changes sign at the endpoints.

Here is a direct unit-circle feasibility identity. Put

$$\eta = \frac{p_3 - p_0}{p_5}, \qquad \kappa = 1 - \eta, \qquad Q_*(z) = \sqrt{p_0 p_5}\,(1 - \kappa z^2 + \kappa z^3 - z^5). \tag{9}$$

The expressions are well-defined since $p_0 p_5 > 0$. One has, for $h = 0, \dots, 5$,

$$\sum_{k=0}^{5-h} \bigl(p_k p_{k+h} + q_k q_{k+h}\bigr) = \begin{cases} 1, & h = 0, \\ 0, & h > 0, \end{cases} \tag{10}$$

where $Q_*(z) = \sum_{k=0}^{5} q_k z^k$. Indeed, $q_k q_j = p_0 p_5 v_k v_j$ with $v = (1, 0, -\kappa, \kappa, 0, -1)^T$; each of the six identities is an identity of rational functions of w, and the rational reductions in the certificate check them exactly, without any square-root approximations. Equating Fourier coefficients in (10) gives

$$|P_*(z)|^2 + |Q_*(z)|^2 = 1 \qquad (|z| = 1). \tag{11}$$

Therefore $\|P_*\|_\infty \le 1$ and $\mathcal{J}(P_*) = \Lambda$. Since $P_*(1) = 1$, its norm is exactly one. We have proved

$$S_5 \ge \Lambda. \tag{12}$$

This explicit complementary-square factorization performs two tasks: it verifies primal feasibility and, through its autocorrelation identities, supplies the trace relation needed in the sharp local dual construction.

3 Phase functionals and Toeplitz majorants

For $b = (b_0, \dots, b_5) \in \mathbb{T}^6$ and $P(z) = \sum a_k z^k$, define

$$\Phi_b(P) = b^T a = \sum_{k=0}^{5} b_k a_k, \qquad L(b) = \max_{\|P\|_\infty \le 1} |\Phi_b(P)|, \qquad R_b = \bar{b}\, b^T. \tag{13}$$

Here $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, transpose T does not mean conjugate transpose, and R_b is the Hermitian matrix satisfying $a^* R_b a = |b^T a|^2$. Since $\sum |a_k| = \max_{b \in \mathbb{T}^6} |b^T a|$,

$$S_5 = \max_{b \in \mathbb{T}^6} L(b). \tag{14}$$

Lemma 3.1 (Finite positive Toeplitz moments)**.** *Suppose $T = (t_{k-j})_{j,k=0}^{5}$ is Hermitian Toeplitz and positive semidefinite. There exist finitely many $\zeta_l \in \mathbb{T}$ and $\mu_l \geq 0$ such that*

$$T_{jk} = \sum_l \mu_l \overline{\zeta_l}^{\,j} \zeta_l^k, \qquad \sum_l \mu_l = t_0. \tag{15}$$

Consequently, $a^*Ta \leq t_0 \|P\|_\infty^2$ for the coefficient vector a of any polynomial of degree at most five.

Proof. Choose Gram vectors $v_0, \ldots, v_5$ satisfying $\langle v_j, v_k \rangle = T_{jk}$, with the inner product linear in its second argument. Toeplitzness implies that $v_j \mapsto v_{j+1}$ is a well-defined isometry from $\operatorname{span}(v_0, \ldots, v_4)$ onto $\operatorname{span}(v_1, \ldots, v_5)$: any linear relation is preserved, because the norms of corresponding linear combinations agree. The two subspaces have the same finite dimension. Extending this isometry by an arbitrary isometry between their orthogonal complements produces a unitary operator U on $\operatorname{span}(v_0, \ldots, v_5)$. Then $v_j = U^j v_0$. The spectral decomposition of the finite-dimensional unitary U gives spectral values $\zeta_l \in \mathbb{T}$ and orthogonal spectral projections E_l. Taking $\mu_l = \|E_l v_0\|^2$ proves (15). In particular, $a^*Ta = \sum_l \mu_l |P(\zeta_l)|^2 \leq t_0 \|P\|_\infty^2$. □

Proposition 3.2 (Toeplitz domination)**.** *If T is Hermitian Toeplitz and $T - R_b \succeq 0$, then*

$$L(b)^2 \leq T_{00}. \tag{16}$$

Moreover, $L(b)^2$ equals the minimum of T_{00} over all such matrices T.

Proof. Since $R_b \succeq 0$, the hypothesis implies $T \succeq 0$. By Lemma 3.1, $|\Phi_b(P)|^2 = a^* R_b a \leq a^* T a \leq T_{00} \|P\|_\infty^2$, proving (16). For the reverse equality, extend the coefficient functional Φ_b from the six-dimensional polynomial subspace of $C(\mathbb{T})$ to $C(\mathbb{T})$ without changing its norm, by complex Hahn–Banach. The Riesz representation theorem supplies a complex measure ν of total variation $L(b)$ representing this extension. Set $T_{jk} = L(b) \int_{\mathbb{T}} \overline{z}^{\,j} z^k \, d|\nu|(z)$. This matrix is Hermitian Toeplitz, with $T_{00} = L(b)^2$, and Cauchy–Schwarz for $|\nu|$ gives $|\Phi_b(P)|^2 \leq L(b) \int |P|^2 d|\nu| = a^* T a$. Thus $T - R_b \succeq 0$. (The case $L(b) = 0$ is immediate.) □

The symmetries

$$b_k \mapsto \gamma \omega^k b_k, \quad b \mapsto \overline{b}, \quad (b_0, \ldots, b_5) \mapsto (b_5, \ldots, b_0), \qquad |\gamma| = |\omega| = 1, \tag{17}$$

preserve $L(b)$, by scalar multiplication, rotation of the variable, conjugation, and reversal of the polynomial. Choosing γ and a fifth root ω allows $b_0 = b_5 = 1$. We therefore parameterize all normalized phase classes by

$$b(\theta) = (1, e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3}, e^{i\theta_4}, 1), \qquad \theta \in (\mathbb{R}/2\pi\mathbb{Z})^4. \tag{18}$$

The normalized phases associated with the orbit of P_* and its reversal are precisely the ten points

$$(0,0,0,\pi) + \frac{2\pi r}{5}(1,2,3,4), \quad (\pi,0,0,0) + \frac{2\pi r}{5}(1,2,3,4), \qquad 0 \leq r \leq 4, \tag{19}$$

with angles taken modulo 2π.

4 A locally sharp continuous Toeplitz certificate

This section proves an exact, not approximate, phase bound near each point in (19). Several small algebraic inequalities are discharged by the accompanying certificate, using fully specified matrices. We distinguish these inputs from the analytic perturbation argument.

4.1 The sharp base point and its nullspace

Let $p = (p_k)_{k=0}^5$, $q = (q_k)_{k=0}^5$, and s be as in (3), (9), and (8). Write

$$N = [p\ q] \in \mathbb{R}^{6\times 2}, \qquad \Pi = N(N^TN)^{-1}N^T. \tag{20}$$

Thus Π is the orthogonal projection onto $\operatorname{span}\{p, q\}$, provided N^TN is invertible, as certified below. Define $t_0 = \Lambda^2$ and t_1, t_2 by

$$At_1 + Bt_2 = \Lambda - t_0\{p_0 - \kappa p_3 + (1-\kappa^2)p_5\}, \tag{21}$$
$$Ct_1 + Dt_2 = \Lambda - t_0\{(1-\kappa^2)p_0 - \kappa p_2 + p_5\}, \tag{22}$$

where

$$A = p_1 + \kappa p_3 + \eta p_4 + \kappa^2 p_5, \qquad B = p_2 + p_3 + \kappa p_4,$$
$$C = \kappa^2 p_0 + \eta p_1 + \kappa p_2 + p_4, \qquad D = \kappa p_1 + p_2 + p_3.$$

Finally set

$$t_3 = t_2 - \kappa t_0 + \kappa t_1, \qquad t_4 = \eta t_1 + \kappa t_2, \qquad t_5 = (1-\kappa^2)t_0 + \kappa^2 t_1, \tag{23}$$

and let $T_* = (t_{|j-k|})_{j,k=0}^5$. The determinant of the system (21)–(22) is nonzero; it is certified by rational interval arithmetic in $\mathbb{Q}(w)$.

Lemma 4.1 (Base-point algebraic certificate)**.** *With $S_0 = T_* - ss^T$, one has*

$$T_*p = \Lambda s, \qquad T_*q = 0, \qquad S_0N = 0, \qquad S_0 + 5\Pi - 4I_6 \succ 0. \tag{24}$$

Consequently $S_0 \succeq 0$, $\ker S_0 = \operatorname{ran} N$, and its restriction to $(\operatorname{ran} N)^\perp$ is strictly greater than $4I$.

Proof. The first two identities reduce to exact identities in $\mathbb{Q}(w)$ after using (21)–(23); the quartic-field part of the certificate checks all twelve components. Moreover $s^Tp = \Lambda$ and $s^Tq = 0$, so these identities imply $S_0N = 0$. The certificate checks $N^TN \succ 0$ and the last matrix inequality in (24) by outward rational intervals and positive LDL^T pivots. Since S_0 is symmetric, it preserves the decomposition $\operatorname{ran} N \oplus (\operatorname{ran} N)^\perp$. On the first summand it is zero, while on the second $\Pi = 0$ and the last inequality gives $S_0 > 4I$. This proves all claims. □

In particular, $T_* - R_s \succeq 0$ is a sharp dual certificate for Φ_s, because $\Phi_s(P_*) = \Lambda$ and $(T_*)_{00} = \Lambda^2$. The matrix S_0 has an unavoidable two-dimensional nullspace. Put

$$G = (S_0 + \Pi)^{-1} - \Pi. \tag{25}$$

Then G is the inverse of S_0 on $(\operatorname{ran} N)^\perp$, extended by zero on $\operatorname{ran} N$, and $\|G\|_{\mathrm{op}} < 1/4$. By (10) at $h = 0$, $\|N\|_{\mathrm{op}} \le \|N\|_F = 1$.

4.2 First-order correction and a strict quadratic form

Let $\theta_0 = \theta_5 = 0$ and $\|\theta\|_2 \le 1/125$. Put $x_i = \sin\theta_i$ for $1 \le i \le 4$, $r = \|x\|_2$, and $\widehat{x} = (0, x_1, \dots, x_4, 0)^T$. We write $u \odot v$ for componentwise multiplication and define

$$v = s \odot \widehat{x}, \qquad w_2 = s \odot \widehat{x}^{\,2}, \qquad R_1 = sv^T - vs^T, \qquad R_2 = vv^T - \tfrac{1}{2}(sw_2^T + w_2s^T). \tag{26}$$

The power $\widehat{x}^{\,2}$ is componentwise. Let J_h $(1 \le h \le 5)$ be the real skew-symmetric Toeplitz matrix with $+1$ on the hth upper diagonal and -1 on the hth lower diagonal. Let E_h be the real symmetric Toeplitz matrix with $+1$ on both such diagonals and zero elsewhere.

For each x, choose the unique $A_1(x) = \sum_{h=1}^{5} \alpha_h(x) J_h$ minimizing

$$\operatorname{tr}\big(((A_1 - R_1)N)^T G((A_1 - R_1)N)\big) \tag{27}$$

subject to

$$p^T (A_1 - R_1) q = 0. \tag{28}$$

The five coefficients are linear in x. Indeed, the objective in (27) is a nonnegative quadratic form, and the bordered Karush–Kuhn–Tucker system below is nonsingular by a certified exact interval test. Its solution is the unique constrained minimizer: any other minimizer would give a nonzero vector in the kernel of this bordered system by subtracting the two stationary equations. For an explicit, reproducible definition, put

$$c_h = p^T J_h q, \qquad M_{h\ell} = \operatorname{tr}\big((J_h N)^T G(J_\ell N)\big), \qquad \mathcal{B} = \begin{pmatrix} M & c \\ c^T & 0 \end{pmatrix}. \tag{29}$$

For $v = s \odot \widehat{x}$, let $r_h(x) = \operatorname{tr}((J_h N)^T G(R_1 N))$ and $d(x) = p^T R_1 q = \Lambda v^T q$. Then $\binom{\alpha(x)}{\tau(x)} = \mathcal{B}^{-1}\binom{r(x)}{d(x)}$ are precisely the constrained normal equations; $\mathcal{B}$ is verified invertible by the certificate. Define $K(x) = A_1(x) - R_1$, and write

$$K(x) = \sum_{i=1}^{4} x_i K_i. \tag{30}$$

As K is skew-symmetric, $N^T K N = 0$ follows from (28).

Define the real symmetric 4×4 matrix H by its quadratic form

$$x^T H x = \Lambda \sum_{i=1}^{4} |p_i| x_i^2 - (p^T v)^2 - (q^T v)^2 - \operatorname{tr}\big((KN)^T G(KN)\big). \tag{31}$$

This is the decisive nondegeneracy condition. The certificate verifies, by outward rational intervals,

$$H \succ \frac{3}{25} I_4, \qquad \sum_{i=1}^{4} \|K_i\|_F^2 < 100. \tag{32}$$

It follows from Cauchy–Schwarz that $\|K(x)\|_{\mathrm{op}} \le 10r$.

4.3 Second- and third-order corrections

Let $T_2(x) = \sum_{h=1}^{5} \beta_h(x) E_h$ be the real symmetric Toeplitz matrix with zero diagonal satisfying

$$N^T (T_2 - R_2) N = (KN)^T G(KN) + \tfrac{1}{2}(x^T H x) I_2. \tag{33}$$

We specify it uniquely by requiring β to be the minimum Euclidean-norm solution of the $(1,1)$ and $(1,2)$ entries of (33). Concretely, let

$$F = \begin{pmatrix} (p^T E_h p)_{h=1}^{5} \\ (p^T E_h q)_{h=1}^{5} \end{pmatrix}; \tag{34}$$

Define

$$D_2 = N^T R_2 N + (KN)^T G(KN) + \tfrac{1}{2}(x^T H x) I_2, \qquad d_2 = \big((D_2)_{11}, (D_2)_{12}\big)^T.$$

Then $\beta = F^T (F F^T)^{-1} d_2$; the certificate checks $F F^T \succ 0$. In particular, this definition uses only the already specified algebraic data, and no free choice of a second-order correction remains.

It is important that the third entry in the 2×2 symmetric equation is not an extra assumption. From (10), for every zero-diagonal real symmetric Toeplitz matrix U,

$$p^T U p + q^T U q = 0. \tag{35}$$

On the other hand, (31) and $s^T p = \Lambda$, $s^T q = 0$ show

$$\operatorname{tr}\left[N^T R_2 N + (KN)^T G (KN) + \tfrac{1}{2}(x^T H x) I_2\right] = 0. \tag{36}$$

Thus the $(2,2)$ entry follows from the other two, and (33) is consistent. Write

$$S_2(x) = T_2(x) - R_2(x) = \sum_{i,j=1}^{4} x_i x_j S_{ij}, \qquad S_{ij} = S_{ji}. \tag{37}$$

The program certifies

$$\sum_{i,j=1}^{4} \|S_{ij}\|_F^2 < 289, \qquad \text{hence} \qquad \|S_2(x)\|_{\mathrm{op}} \le 17 r^2. \tag{38}$$

For the cubic correction set $e_2 = -w_2/2$ and

$$R_3 = e_2 v^T - v e_2^T. \tag{39}$$

Let

$$J_* = \frac{\sum_{h=1}^{5} c_h J_h}{\sum_{h=1}^{5} c_h^2}, \qquad p^T J_* q = 1, \tag{40}$$

where c_h are as in (29); the denominator is positive by the invertibility checks. Define

$$\gamma_3(x) = p^T\big(R_3 + KGS_2 + S_2GK + KGKGK\big)q, \qquad A_3(x) = \gamma_3(x) J_*, \qquad K_3 = A_3 - R_3. \tag{41}$$

All matrix products are ordinary finite products. By skew-symmetry, the matrix

$$N^T\big(K_3 - KGS_2 - S_2GK - KGKGK\big)N \tag{42}$$

is real skew-symmetric; its only independent entry, the $(1,2)$ entry, vanishes by (41). Hence (42) is identically zero. The certificate obtains the homogeneous symmetric tensor $K_3(x) = \sum_{i,j,k} x_i x_j x_k C_{ijk}$ by polarization and verifies

$$\sum_{i,j,k=1}^{4} \|C_{ijk}\|_F^2 < 16, \qquad \text{hence} \qquad \|K_3(x)\|_{\mathrm{op}} \le 4 r^3. \tag{43}$$

4.4 Schur complement and explicit remainder

Let

$$T(x) = T_* + iA_1(x) + T_2(x) + iA_3(x). \tag{44}$$

Each correction is Hermitian Toeplitz (because i times a real skew-symmetric Toeplitz matrix is Hermitian), and each has zero diagonal, so $T(x)_{00} = \Lambda^2$.

For $b = s \odot e^{i\theta}$, the exact identities

$$\cos\theta_j - 1 = -\frac{x_j^2}{1+\cos\theta_j}, \qquad \cos\theta_j - 1 + \frac{x_j^2}{2} = -\frac{x_j^4}{2(1+\cos\theta_j)^2}$$

show that

$$R_b = ss^T + iR_1 + R_2 + iR_3 + \Delta, \qquad \|\Delta\|_{\mathrm{op}} \le 4r^4 \tag{45}$$

for $r \le 1/125$. Here is an explicit remainder calculation. Set $u = -w_2/2$ and $e = s \odot (\cos\theta - \mathbf{1} + \widehat{x}^{\,2}/2)$, where $\cos\theta$ has entries 1 at indices $0, 5$. Since $\cos\theta_j > 1/2$ and $\sum_j x_j^8 \le (\sum_j x_j^2)^4 = r^8$, the second displayed identity gives

$$\|u\|_2 \le r^2/2, \qquad \|e\|_2 \le r^4/4, \qquad \|v\|_2 \le r, \qquad \|s\|_2 = \sqrt{6}.$$

Now $b = s + u + e + iv$ and $\overline{b} = s + u + e - iv$, so the contributions through total degree three are exactly those displayed in R_1, R_2, R_3. The remaining terms are

$$\Delta = uu^T + (s + u - iv)e^T + e(s + u + iv)^T + ee^T.$$

The rank-one operator-norm identity $\|ab^T\|_{\mathrm{op}} = \|a\|_2\|b\|_2$ therefore yields

$$\|\Delta\|_{\mathrm{op}} \le \frac{r^4}{4} + 2\big(\sqrt{6} + r + r^2/2\big)\frac{r^4}{4} + \frac{r^8}{16} < 4r^4 \qquad (0 < r \le 1/125).$$

At $r = 0$ the remainder is zero, so (45) holds throughout the stated ball.

Consequently

$$T(x) - R_b = S_0 + iK + S_2 + iK_3 - \Delta. \tag{46}$$

We analyze this Hermitian matrix relative to the orthogonal decomposition $\operatorname{ran} N \oplus (\operatorname{ran} N)^{\perp}$. Let

$$h(r) = 17r^2 + 4r^3 + 4r^4, \quad e(r) = 10r + h(r), \quad g(r) = 4r^3 + 4r^4. \tag{47}$$

By Lemma 4.1 and (32)–(45), the lower-right (range) block is bounded below by $(4 - e(r))I$. Its inverse has norm at most $1/(4 - e(r))$. Two applications of the resolvent identity, expanding around the inverse G of S_0 and retaining the constant and linear terms, bound the residual inverse by

$$\frac{h(r)}{16} + \frac{e(r)^2}{16(4 - e(r))}. \tag{48}$$

Indeed, if D is the range perturbation with $\|D\| \le e(r)$ and D_1 its linear term with $\|D - D_1\| \le h(r)$, the identity

$$(S_0 + D)^{-1} = G - GDG + GDGD(S_0 + D)^{-1}$$

on the range gives (48), using $\|G\| \le 1/4$ and replacing D by D_1 in the first-order term.

To make the Schur remainder exhaustive, write

$$A = iK, \quad S = S_2, \quad Z = iK_3 - \Delta, \quad B = A + S + Z, \quad V = S + Z.$$

Then $\|A\| \le 10r$, $\|Z\| \le g(r)$, $\|V\| \le h(r)$, and $\|B\| \le e(r)$. Let $\mathsf{P} = I_6 - \Pi$ be the orthogonal projection onto $\operatorname{ran}(N)^{\perp}$. The range-block perturbation is $D = \mathsf{P}B\mathsf{P}$ (restricted to that range), and its linear part is $D_1 = \mathsf{P}A\mathsf{P}$. With G viewed as zero on $\operatorname{ran}(N)$, write

$$(S_0|_{\operatorname{ran}\mathsf{P}} + D)^{-1} = G - GD_1G + \mathcal{R}, \qquad \|\mathcal{R}\| \le \frac{h(r)}{16} + \frac{e(r)^2}{16[4 - e(r)]}.$$

Indeed, twice applying the resolvent identity gives $G - GDG + GDGD(S_0 + D)^{-1}$, and replacing D by D_1 in the linear term introduces an error bounded by $h(r)/16$.

The Schur complement, expressed in the two-dimensional coordinate basis N (which need not be orthonormal), is

$$\mathcal{S} = N^T BN - N^T B(G - GD_1G + \mathcal{R})BN.$$

Because $N^T AN = 0$, its homogeneous terms of orders two and three are respectively

$$N^T SN - N^T AGAN = \tfrac{1}{2}(x^T Hx)I_2$$

and

$$N^T\big(iK_3 - AGS - SGA + AGAGA\big)N = iN^T\big(K_3 - KGS - SGK - KGKGK\big)N = 0.$$

We used $A = iK$ and (33), (42). The remaining difference has norm at most

$$E(r) = 4r^4 + \frac{20r\,g(r) + h(r)^2}{4} + \frac{10r}{16}\big(20r\,h(r) + h(r)^2\big) + [10r + h(r)]^2\left(\frac{h(r)}{16} + \frac{e(r)^2}{16[4 - e(r)]}\right). \tag{49}$$

For completeness, the four terms follow by estimating, in order, $\|N^T \Delta N\| \le 4r^4$;

$$\|BGB - AGA - AGS - SGA\| \le \frac{2(10r)g(r) + h(r)^2}{4};$$

$$\|BGD_1GB - AGD_1GA\| \le \frac{10r}{16}\big(2(10r)h(r) + h(r)^2\big);$$

and $\|B\mathcal{R}B\| \le e(r)^2\|\mathcal{R}\|$. Every displayed estimate uses only submultiplicativity, $\|G\| < 1/4$, $\|N\| \le 1$ and the preceding bounds. In these products G annihilates the nullspace, so the range projections may be suppressed without changing the expressions.

The certificate checks as rational inequalities, for $R = 1/125$,

$$e(R) < 4, \qquad E(R) < \frac{3}{50}R^2. \tag{50}$$

Every coefficient in $E(r)$ and $e(r)$ is nonnegative, and $4 - e(r)$ decreases on $[0, R]$; therefore $E(r)/r^2$ is nondecreasing there. Hence (50) applies to all $0 < r \le R$. By (32), the main Schur term is greater than $(3/50)r^2 I_2$, while its error is smaller than that. Both the range block and the Schur complement are strictly positive. Thus $T(x) - R_b \succ 0$ for $0 < r \le 1/125$; at $r = 0$ it equals $S_0 \succeq 0$.

Theorem 4.2 (Sharp local phase bound)**.** *For $\|\theta\|_2 \le 1/125$,*

$$L(1, s_1e^{i\theta_1}, \dots, s_4e^{i\theta_4}, 1) \le \Lambda, \tag{51}$$

with strict inequality unless $\theta = 0$. The same assertion, after the symmetries (17), holds in the corresponding balls around all ten phase centers (19).

Proof. Since $|\theta_j| \le 1/125 < \pi/2$, the equality $r = 0$ is equivalent to $\theta = 0$. Equations (44)–(50) give a Hermitian Toeplitz matrix with diagonal Λ^2 majorizing R_b, strictly when $\theta \ne 0$. Proposition 3.2 gives the desired bound, and strict positive definiteness gives strict inequality for every nonzero polynomial. Symmetries preserve L. □

5 Excluding phase boxes without a global Lipschitz loss

Set

$$L_0 = \frac{21865297718925}{10^{13}} = 2.1865297718925 < \Lambda. \tag{52}$$

The strict comparison is proved by the rational enclosure for w in the certificate.

Lemma 5.1 (Phase-aligned box exclusion)**.** *Let $B(c,h)$ be a coordinate box in the normalized phase torus (18), centered at $c \in (\mathbb{R}/2\pi\mathbb{Z})^4$ and with each coordinate at angular distance at most $h < \pi/2$ from the corresponding coordinate of c. If*

$$L(b(c)) < L_0 \cos h, \tag{53}$$

then $B(c,h)$ cannot contain the phase vector associated with any maximizing degree-five polynomial.

Proof. A maximizing polynomial exists by compactness of the unit ball in the finite-dimensional coefficient space. For its coefficient vector a, choose $b_k = \overline{a_k}/|a_k|$ when $a_k \neq 0$ and any unit phase when $a_k = 0$. Normalize $b_0 = b_5 = 1$ using (17), with the corresponding norm-preserving transformation of the polynomial. Suppose the resulting phase vector lies in $B(c,h)$. With $\rho_k = |a_k|$, the center has coefficient phase errors $\delta_0 = \delta_5 = 0$ and $|\delta_k| \le h$ for $1 \le k \le 4$. Hence

$$L(b(c)) \ge \Re\Phi_{b(c)}(P) = \sum_{k=0}^{5} \rho_k \cos\delta_k \ge \cos h \sum_{k=0}^{5} \rho_k = \cos h\, S_5 \ge \cos h\, \Lambda > L(b(c)),$$

a contradiction. This statement concerns *globally maximizing aligned phases*; it does not assert $L(b(\theta)) \le L_0$ for every θ in the box. □

Two convenient sufficient tests for (53) follow from Proposition 3.2 and Fourier interpolation.

Lemma 5.2 (Five-point Fourier test)**.** *For $b = (1, b_1, b_2, b_3, b_4, 1)$, let $\zeta = e^{2\pi i/5}$ and*

$$d_j = \frac{1}{5} \sum_{k=0}^{4} b_k \zeta^{-jk} \quad (0 \le j \le 4). \tag{54}$$

Then $L(b) \le \sum_{j=0}^{4} |d_j|$.

Proof. By discrete Fourier inversion, $\sum_j d_j \zeta^{jk} = b_k$ for $k = 0, \dots, 4$. For $k = 5$, the left side equals $\sum_j d_j = b_0 = b_5 = 1$. Thus $\Phi_b(P) = \sum_j d_j P(\zeta^j)$, giving the bound. □

6 A complete finite covering of the phase torus

The certificate described here verifies a finite exact assertion. We specify every class of leaf, the recursive partition, the arithmetic model, the counts, and the connection to Lemma 5.1. In particular, the reader does not need to trust a numerical optimizer.

6.1 Initial cells and exact symmetry reduction

At level $\ell = 0$, take all boxes with centers

$$c = \frac{2\pi}{32}(t_1, t_2, t_3, t_4), \qquad t_j \in \{0, 1, \ldots, 31\}, \qquad h_0 = \frac{\pi}{32}.$$

They cover the full normalized four-torus, including boundary points. Conjugation and reversal generate the transformations $t \mapsto -t$ and $t \mapsto (t_4, t_3, t_2, t_1)$ on this grid. The group has order four. Burnside's lemma gives

$$\frac{32^4 + 32^2 + 32^2 + 16}{4} = 262660 \tag{55}$$

representative boxes, selecting the lexicographically least tuple in each orbit. These symmetries preserve the problem and map each box to another box, so examining representatives suffices.

At level $\ell \geq 0$, put

$$h_\ell = \frac{\pi}{32 \cdot 2^\ell}, \qquad n_0 = 32, \quad n_\ell = 32 \cdot 2^{\ell+1} \ (\ell \geq 1), \tag{56}$$

and represent each center by an integer tuple t with $c = 2\pi t / n_\ell$. A nonterminal level-zero box has exactly sixteen children with center tuples $4t + (\pm 1, \pm 1, \pm 1, \pm 1)$ modulo 128. At all later levels the children have tuples $2t + (\pm 1, \pm 1, \pm 1, \pm 1)$ modulo $n_{\ell+1}$. This subdivision gives sixteen closed subboxes of half the original side length, with no gaps.

6.2 Four verified node types

Each visited box is assigned exactly one of the following four types.

1. **Fourier leaf.** Using (54) at the center, the certificate proves $\sum_j |d_j| < L_0 \cos h_\ell$ by outward interval arithmetic. Lemmas 5.2 and 5.1 exclude the entire box.

2. **Local leaf.** The certificate proves that the Euclidean angular distance from the box center to one of the ten points (19), plus $2h_\ell$ (the Euclidean half-diagonal of the box), is strictly less than $1/125$. The entire box is then contained in a ball treated by Theorem 4.2.

3. **Toeplitz leaf.** The certificate provides five complex rational moments $m_1, \ldots, m_5$, with real and imaginary parts of denominator 10^{12}, and defines the Hermitian Toeplitz matrix T by $T_{00} = (L_0 \cos h_\ell)^2$, $T_{j,k} = m_{k-j}$ for $j < k$, and conjugate symmetry. Outward interval LDL^* elimination proves $T - R_{b(c)} \succ 0$. Proposition 3.2 and Lemma 5.1 exclude the box.

4. **Subdivision node.** The program visits all sixteen children as specified above.

These alternatives are exhaustive: every node, starting from each of the (55) representatives, has exactly one recorded type; every subdivided node creates every one of its sixteen children. The verifier rejects missing or surplus records.

6.3 Certified counts and integer arithmetic

Proposition 6.1 (Finite phase-cover certificate)**.** *The accompanying finite verifier passes all tests described above with the following exact counts:*

Initial symmetry-orbit representatives	*262660*
All visited nodes	*272852*
Fourier leaves	*248376*
Local leaves	*549*
Toeplitz leaves	*23290*
Subdivision nodes	*637*
Maximum leaf level	*7*

The subdivision counts at levels $0, \dots, 6$ *are respectively* $94, 93, 101, 101, 101, 102, 45$. *All interval* LDL^* *pivots are strictly positive; their smallest certified lower endpoint is*

$$\frac{3505258205231374557056651943921957879845929}{10^{50}} > 0.$$

The 2,136,052-byte uncompressed node-and-moment payload has SHA-256 digest

`5c9c8ad7091a626d37e0dfde3694d7e30c751f94c5ae52326c9e51180c9e1577`.

Proof. This is a finite exact computation. Appendix A specifies its decoding, arithmetic, coverage recursion, and verification conditions; `Sidon5_Exact_Closure_Certificate_Revised.py` implements them verbatim. The digest is an integrity check, not an alternative to the mathematical inequalities: all intervals, matrices, and child relations are recalculated and verified by the program. The recorded counts also satisfy $272852 = 262660 + 16 \cdot 637$, as required by exhaustive subdivision. □

Theorem 6.2 (Global phase exclusion)**.** *If* P *is a global maximizing polynomial for* S_5*, then its normalized aligned coefficient phases lie at one of the ten points* (19)*. Moreover* $S_5 = \Lambda$.

Proof. Normalize the aligned phase vector as in Lemma 5.1, and choose an initial grid box containing it. By symmetry, we may work with its representative. Proposition 6.1 recursively partitions this box until a verified leaf is reached. Fourier or Toeplitz leaves cannot contain the phase vector of a global maximizer, by Lemma 5.1; hence the phase lies in a local leaf, and so in the neighborhood of Theorem 4.2. Thus its phase-functional norm is at most Λ, and by alignment $S_5 = |\Phi_b(P)| \le L(b) \le \Lambda$. Combining with (12) proves equality. The strict part of Theorem 4.2 then forces the normalized phase to equal the corresponding center. □

7 Uniqueness and the full equality classification

It remains to determine the maximizing polynomial for a fixed optimal phase functional. By the symmetries (17), it is enough to treat $s = (1, 1, 1, 1, -1, 1)$.

Proposition 7.1 (Uniqueness at the sharp phase)**.** *If* $\|P\|_\infty \le 1$ *and* $\Phi_s(P) = \Lambda$*, then* $P = P_*$. *More generally, if* $|\Phi_s(P)| = \Lambda$*, then* P *is a unimodular multiple of* P_*.

Proof. The Toeplitz inequalities and $|\Phi_s(P)| = \Lambda$ yield

$$\Lambda^2 = |\Phi_s(P)|^2 \le a^* T_* a \le \Lambda^2 \|P\|_\infty^2 \le \Lambda^2.$$

If $\Phi_s(P) = \Lambda$, equality forces $a^* S_0 a = 0$, so Lemma 4.1 gives $a \in \ker_{\mathbb{C}} S_0 = \operatorname{span}_{\mathbb{C}}\{p, q\}$. Since $s^T p = \Lambda$ and $s^T q = 0$, the equation $s^T a = \Lambda$ implies

$$P = P_* + \beta Q_*, \qquad \beta \in \mathbb{C}. \tag{57}$$

We show $\beta = 0$ from the necessary angular derivative conditions at two contact points. By (11), at every zero z of Q_* on $\mathbb{T}$ we have $|P_*(z)| = 1$. Also $Q_*(1) = 0$ and $P_*(1) = 1$. Hence any feasible polynomial (57) has modulus one at $z = 1$, and its angular derivative of squared modulus must vanish there. As P_* has real coefficients,

$$0 = \frac{d}{dt}|P(e^{it})|^2\Big|_{t=0} = 2\,\mathrm{Re}\big(i(P_*'(1) + \beta Q_*'(1))\big).$$

Now $Q_*'(1) = (\kappa - 5)\sqrt{p_0 p_5} \in \mathbb{R} \setminus \{0\}$, forcing $\mathrm{Im}\,\beta = 0$.

The complementary polynomial factors as

$$Q_*(z) = \sqrt{p_0 p_5}\,(1 - z)(1 + z + \eta z^2 + z^3 + z^4). \tag{58}$$

The bound $0.834 < \eta < 0.835$ follows by rational isolation. For a unit-circle zero of the quartic, divide its equation by z^2 and set $x = \cos t$ to obtain $4x^2 + 2x + \eta - 2 = 0$. Its two roots

$$c = \frac{-1 + \sqrt{9 - 4\eta}}{4}, \qquad d = \frac{-1 - \sqrt{9 - 4\eta}}{4} \tag{59}$$

are distinct and belong to $(-1, 1)$; hence Q_* has five distinct unit-circle zeros, including 1. Choose $z = c + i\sqrt{1 - c^2}$. The same derivative condition at this contact point, now with $\beta \in \mathbb{R}$, gives

$$0 = 2\beta\,\mathrm{Re}\left(\overline{P_*(z)}\,iz Q_*'(z)\right). \tag{60}$$

The certificate evaluates the algebraic expression in parentheses with rigorous rational intervals and proves it is nonzero. Thus $\beta = 0$. For arbitrary complex value $\Phi_s(P)$ of modulus Λ, multiply P by a scalar of modulus one to make this value Λ and apply the result. □

Proof of Theorem 1.1. Section 2 gives the attainable lower bound Λ. Theorem 6.2 supplies the matching global upper bound and restricts equality phases to (19). Proposition 7.1 classifies the norming polynomial at one center. Undoing scalar and variable rotations and coefficient reversal gives exactly (7). Conversely, every polynomial in (7) has supremum norm one and coefficient modulus sum Λ, by (11). The real polynomial P_* attains the complex optimum, so the real and complex Sidon constants agree. □

8 Interpretation and further questions

The fundamental obstacle in transferring the cubic method to degree five is not the algebraic construction of an almost-flat extremizer: (11) already settles its feasibility. The obstruction is the four-dimensional continuum of coefficient phases. A sharp Toeplitz matrix at the candidate necessarily loses strict positivity along two coefficient directions, so a naive perturbation of the base matrix cannot uniformly prove the sharp inequality. The explicit cubic correction removes precisely the resulting third-order Schur obstruction; the positive quadratic form H then furnishes a genuine open neighborhood of sharp control.

The second observation is that one need not prove the sharp inequality by a single formula valid over the entire phase torus. At a *maximizer*, the coefficient phases align with the maximizing functional. Lemma 5.1 turns this alignment into a strict exclusion criterion for cells, leaving only finitely many neighborhoods in which sharpness matters. The finite verifier checks the exhaustive partition and the matrix bounds, rather than extrapolating from a numerical grid. This distinction

is essential: a finite mesh with an error factor greater than one would not itself establish the exact constant.

The proof establishes the global value and the complete equality classification, but it leaves open an interesting question of proof economy: can the thousands of off-center rational matrices be replaced by a short universal analytic Toeplitz family, retaining the sharp local construction? More generally, the local-sharp/global-exclusion scheme may be useful for Sidon constants of other finite spectra, provided an algebraic extremizer and an isolatable family of phase centers are available. No such extension is claimed here.

A The independent finite verifier

The sole proof-bearing program is `Sidon5_Exact_Closure_Certificate_Revised.py`. It is distributed beside this LaTeX source and runs with the command

```
python3 Sidon5_Exact_Closure_Certificate_Revised.py
```

without a solver, third-party libraries, network access, or prior computed files. Assertions are part of the verification, so Python must not be run with `-O`. All real intervals are represented by two signed arbitrary-precision integers $[L, U]$ corresponding to $[L/10^{50}, U/10^{50}]$. Addition, subtraction, multiplication, positive division, square, and nonnegative square root round their endpoints outward; the last operation uses the integer square root. Complex intervals are pairs of real intervals.

The analytic inputs passed to the verifier are precisely these:

1. Arithmetic in $\mathbb{Q}[w]/(16w^4+10w^3-8w+1)$ proves (6), (10), $P_*(1) = 1$, the sign pattern, $T_*p = \Lambda s$, and $T_*q = 0$. Rational root isolation bounds the associated algebraic coefficients.

2. Real LDL^T tests verify $N^TN \succ 0$, the gap (24), invertibility of $\mathcal{B}$ and FF^T, and the positive quadratic matrix (32). The code constructs the minimizer and the polarization tensors from (27)–(41), then verifies (32), (38), (43), and (50), using rational intervals.

3. The program reconstructs π from $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ with alternating Taylor bounds (50 terms for each arctangent), and roots of unity by repeated half-angle square roots and complex multiplication. The identity for π follows from the tangent addition formula and the fact that $0 < 4\arctan(1/5) - \arctan(1/239) < \pi/2$ has tangent one. Thus every trigonometric enclosure is rigorous.

4. The global payload is decoded from base85 and zlib and checked against the displayed SHA-256 digest. A depth-first traversal of each of the 262660 initial orbit representatives reads exactly one tag per node and, for Toeplitz leaves, ten signed 64-bit integers representing five complex moments. At the end the byte stream must be exhausted and all expected counts must match. Every Fourier, local and Toeplitz leaf is independently rechecked against the corresponding strict inequality. Toeplitz positive definiteness is proved by interval complex LDL^*, requiring the lower endpoint of each pivot to be positive.

For avoidance of ambiguity, the recursion implemented in the file has the following mathematical pseudocode. All tests are *strict* comparisons of enclosing intervals; failure at any assertion invalidates the certificate.

```
VERIFY-BOX(level, t):
  read next tag; reject missing or invalid tag
  if tag = FOURIER:
```

```
    compute interval enclosure of DFT mass at center
    assert upper(mass) < lower(L0 cos(h[level]))
  else if tag = LOCAL:
    assert a ten-center ball contains the complete box
  else if tag = TOEPLITZ:
    read ten integers; construct the six-by-six Hermitian
    Toeplitz matrix with diagonal (L0 cos(h[level])) squared
    assert every interval LDL-star pivot of T-R(center) is positive
  else if tag = SPLIT:
    for all 16 sign vectors with entries -1 or +1:
      VERIFY-BOX(level+1, exact-child(level,t,sign))
  else:  reject
FOR each lexicographic initial orbit representative t:
  VERIFY-BOX(0,t)
assert all bytes consumed, node counts and layer counts exact.
```

For a Hermitian matrix A, the interval LDL^* recurrence uses

$$d_j = A_{jj} - \sum_{k<j} |L_{jk}|^2 d_k, \qquad L_{ij} = \frac{A_{ij} - \sum_{k<j} L_{ik}\overline{L_{jk}} d_k}{d_j},$$

where each positive divisor has a strictly positive certified lower bound. By induction, the true pivots are contained in the computed intervals. Six positive pivots certify $A \succ 0$, with no floating-point eigenvalue approximation. The minimum reported pivot is an LDL^* pivot lower bound, not a uniform lower bound for all minimum eigenvalues.

The payload is compressed solely for file size: decoding is deterministic, and the verifier does not accept its recorded labels or moments without recomputing all associated inequalities. In particular, the digest authenticates one dataset but is not itself a proof of the mathematical bounds. The proof obligation is jointly discharged by the arguments of Sections 3–6 and the full execution of the independent verifier.

References

[1] F. Bach and A. Rudi, "Exponential convergence of sum-of-squares hierarchies for trigonometric polynomials," *SIAM Journal on Optimization* **33** (2023), no. 3, 2137–2159, doi:10.1137/22M1540818.

[2] E. Bombieri and J. Bourgain, "On Kahane's ultraflat polynomials," *Journal of the European Mathematical Society* **11** (2009), no. 3, 627–703, doi:10.4171/JEMS/163.

[3] E. Christensen, "The Schur multiplier norm and its dual norm," *Linear Algebra and its Applications* **743** (2026), 87–101, doi:10.1016/j.laa.2026.04.022.

[4] J.-P. Kahane, "Sur les polynômes à coefficients unimodulaires," *Bulletin of the London Mathematical Society* **12** (1980), no. 5, 321–342, doi:10.1112/blms/12.5.321.

[5] S. Neuwirth, "On the (Fourier analytic) Sidon constant of $\{0, 1, 2, 3\}$," arXiv:2603.28229 (2026; circulation of an earlier lecture manuscript).

[6] H. Yu, "The exact Sidon constant of $\{0, 1, 2, 3\}$ and its formalization," viXra:2606.0009v1 (2026).